

Séminaire de Géométrie Algébrique du Bois Marie
1960–61

Étale coverings and the fundamental group (SGA 1)

Preface, Introduction, Foreword

English draft

A seminar directed by A. Grothendieck,
augmented by two exposés of Mme M. Raynaud

About this draft

This is an unofficial English translation of the front matter of A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61: the abstract, the preface of the SMF edition, Grothendieck’s Introduction (Massy, August 1970), and the Foreword (*Avertissement*, Bures, June 1963) to the mimeographed notes. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2. The exposés themselves are in the sibling directories **ExposeI**, . . . , **ExposeXIII**. Scholarly proofreading remains outstanding.

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Abstract

This volume is a newly typeset and annotated edition of the book “Revêtements Étales et Groupe Fondamental”, Lecture Notes in Mathematics, 224, Springer-Verlag, Berlin-Heidelberg-New York, 1971, by Alexander Grothendieck et al.

The text presents the foundations of a theory of the fundamental group in Algebraic Geometry, from the “Kroneckerian” point of view, which makes it possible to treat on the same footing the case of an algebraic variety in the usual sense and that of the ring of integers of a number field, for example.

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Preface

The present text is a newly typeset and annotated edition of the book “Revêtements Étales et Groupe Fondamental”, Lecture Notes in Mathematics, 224, Springer-Verlag, Berlin-Heidelberg-New York, 1971, by Alexander Grothendieck et al.

The typesetting in L^AT_EX2e was carried out by volunteers participating in a project directed by Bas Edixhoven, about which more details can be found at the address <http://www.math.leidenuniv.nl/~edix/>. The page layout was completed by the Société mathématique de France.

This is a *slightly corrected version* of the original text. It is published by the Société mathématique de France (<http://smf.emath.fr/>) in the series “Documents Mathématiques”. Some updates have been made by Michel Raynaud. These are delimited by brackets [] and indicated by the symbol (MR) (Remarks X 2.14, XI 1.4, XII 5.6, XIII 2.13, and the footnote to III 6.6). In order to unify notation, the residue field of a point x is denoted $k(x)$, and that of a local ring A is denoted $k(A)$.

There also exists an electronic version meant to reproduce the original text.

The two versions have a single common source file, which is to be found on the arXiv.org e-Print archive server at <http://arxiv.org/>. The differences between the two versions are documented in the source file.

The old page numbering is incorporated in the margin (the number n marking the beginning of page n).¹

¹Translator’s note: in this English edition the original page numbers are kept only as comments in the T_EX source, and the page references of the SMF edition in the preceding paragraphs have been replaced by statement numbers.

Introduction

In the first part of this introduction we give some details on the contents of the present volume; in the second, on the whole of the “*Séminaire de Géométrie Algébrique du Bois-Marie*”, of which the present volume constitutes the first volume.

1. The present volume presents the foundations of a theory of the fundamental group in Algebraic Geometry, from the “Kroneckerian” point of view, which makes it possible to treat on the same footing the case of an algebraic variety in the usual sense and that of the ring of integers of a number field, for example. This point of view can be expressed in a satisfactory way only in the language of schemes, and we shall use this language freely, as well as the main results expounded in the first three chapters of the *Éléments de Géométrie Algébrique* of J. DIEUDONNÉ and A. GROTHENDIECK (cited EGA in what follows). The study of the present volume of the “*Séminaire de Géométrie Algébrique du Bois-Marie*” requires no other knowledge of Algebraic Geometry, and can therefore serve as an introduction to the present-day techniques of Algebraic Geometry for a reader wishing to become familiar with these techniques.

Exposés I to XI of this book are a verbatim reproduction, practically unchanged, of the mimeographed notes of the oral Seminar, which were distributed by the *Institut des Hautes Études Scientifiques*². We have confined ourselves to adding a few footnotes to the original text, correcting some misprints, and making a terminological adjustment, the term “simple morphism” in particular having been replaced in the meantime by “smooth morphism”, which does not lend itself to the same confusions.

Exposés I to IV present the local notions of *étale* morphism and *smooth* morphism; they make hardly any use of the language of schemes, expounded in Chapter I of the *Éléments*³. Exposé V presents the axiomatic description of the fundamental group of a scheme, useful even in the classical case where this scheme reduces to the spectrum of a field, where one finds a very convenient reformulation of the usual Galois theory. Exposés VI and VIII present the *theory of descent*, which has taken on a growing importance in Algebraic Geometry in recent years, and which could render analogous services in Analytic Geometry and in Topology. It should be pointed out that Exposé VII had not been written up, and its substance is to be found incorporated in a work of J. Giraud (*Méthode de la Descente*, Bull. Soc. Math. France, Mémoire 2, 1964, viii + 150 p.). In Exposé IX one studies more specifically the descent of étale morphisms, obtaining a systematic approach to theorems of VAN KAMPEN type for the fundamental group, which appear here as simple translations of descent theorems. It is essentially a procedure for computing the fundamental group of a connected scheme X endowed with a surjective and proper morphism, say $X' \rightarrow X$, in terms of the fundamental groups of the connected components of X' and of the fibered products $X' \times_X X'$, $X' \times_X X' \times_X X'$, and of the homomorphisms induced between these groups by the canonical simplicial morphisms between the preceding schemes. Exposé X gives the theory of *specialization of the fundamental group* for a proper and smooth morphism, the most striking result of which consists in the determination (to within very little) of the

²As were the notes of the seminars following this one. This mode of distribution having proved impracticable and insufficient in the long run, all the “*Séminaire de Géométrie Algébrique du Bois-Marie*” will henceforth appear in book form, like the present volume.

³A more complete study is now available in the *Éléments*, Chap. IV, §§ 17 and 18.

fundamental group of a smooth algebraic curve in characteristic $p > 0$, thanks to the result known by transcendental means in characteristic zero. Exposé XI gives some examples and complements, making explicit in particular, in cohomological form, the theory of *KUMMER* coverings, and that of *ARTIN-SCHREIER*. For further comments on the text, see the *Foreword* to the mimeographed version, which follows the present Introduction.

Since the writing of the present Seminar in 1961, there have been developed, in collaboration by M. ARTIN and myself, the language of *étale topology* and a corresponding cohomological theory, expounded in the part SGA 4 “Étale cohomology of schemes” of the *Séminaire de Géométrie Algébrique*, to appear in the same series as the present volume. This language, and the results it already has at its disposal, provide a particularly flexible tool for the study of the fundamental group, making it possible to understand better, and to go beyond, some of the results expounded here. There would therefore be reason to take up the theory of the fundamental group entirely anew from this point of view (all the key results in fact already appearing in loc. cit.). This is what was planned for the chapter of the *Éléments* devoted to the fundamental group, which was also to contain several other developments that could not find their place here (relying on the technique of resolution of singularities): computation of the “local fundamental group” of a complete local ring in terms of a suitable resolution of singularities of this ring, local and global Künneth formulas for the fundamental group without a properness hypothesis (cf. Exp. XIII), the results of M. ARTIN on the comparison of the local fundamental groups of an excellent henselian local ring and of its completion (SGA 4 XIX). Let us also point out the necessity of developing a theory of the fundamental group of a topos, which will encompass at the same time the ordinary topological theory, its semi-simplicial version, the “profinite” variant developed in Exposé V of the present volume, and the somewhat more general pro-discrete variant of SGA 3 X 7 (adapted to the case of non-normal and non-unibranch schemes). Pending an overall recasting of the theory from this point of view, Exposé XIII by Mme RAYNAUD, which uses the language and the results of SGA 4, is intended to show the advantage that can be drawn from them in some typical questions, in particular by generalizing certain results of Exposé X to non-proper relative schemes. In it one gives in particular the structure of the “prime-to- p ” fundamental group of a non-complete algebraic curve in arbitrary characteristic (which I had announced in 1959, but of which no proof had been published up to now).

In spite of these numerous gaps and imperfections (others will say: because of these gaps and imperfections), I think that the present volume may be useful to the reader who wishes to become familiar with the theory of the fundamental group, as well as a reference work, pending the writing and publication of a text escaping the criticisms I have just enumerated.

2. The present volume constitutes volume 1 of the “*Séminaire de Géométrie Algébrique du Bois-Marie*”, the following volumes of which are scheduled to appear in the same series as this one. The aim that the *Seminar* sets itself, in parallel with the treatise “*Éléments de Géométrie Algébrique*” of J. DIEUDONNÉ and A. GROTHENDIECK, is to lay the foundations of Algebraic Geometry, following the points of view in this latter work. The standard reference for all the volumes of the *Seminar* is constituted by Chapters I, II, III of the “*Éléments de Géométrie Algébrique*” (cited EGA I, II, III), and the reader is assumed to be in possession of the background in commutative algebra and homological algebra that these chapters presuppose⁴. Moreover, in each volume of the *Seminar* reference will be made freely, to the extent needed, to earlier volumes of the same *Seminar*, or to other chapters of the “*Éléments*” that have been published or are about to appear.

Each *part* of the *Seminar* is centered on a principal subject, indicated in the title of the corresponding volume or volumes; the oral seminar generally extends over one academic year, sometimes more. The exposés within each part of the *Seminar* are generally in close logical dependence on one another; on the other hand, the different parts of the *Seminar* are to a

⁴See the Introduction to EGA I for details on this point.

large extent logically independent of one another. Thus the part “Group Schemes” is almost entirely independent of the two parts of the *Seminar* that precede it chronologically; on the other hand, it makes frequent appeal to the results of EGA IV. Here is the list of the parts of the *Seminar* that are to appear shortly (cited SGA 1 to SGA 7 in what follows):

- SGA 1. Étale coverings and the fundamental group, 1960 and 1961.
- SGA 2. Local cohomology of coherent sheaves and local and global Lefschetz theorems, 1961/62.
- SGA 3. Group schemes, 1963 and 1964 (3 volumes), in coll. with M. DEMAZURE.
- SGA 4. Theory of topoi and étale cohomology of schemes, 1963/64 (3 volumes), (in coll. with M. ARTIN and J.L. VERDIER).
- SGA 5. l -adic cohomology and L -functions, 1964 and 1965 (2 volumes).
- SGA 6. Theory of intersections and the Riemann-Roch theorem, 1966/67 (2 volumes) (in coll. with P. BERTHELOT and L. ILLUSIE).
- SGA 7. Local monodromy groups in algebraic geometry.

Three of these partial *Seminars* were directed in *collaboration* with other mathematicians, who will appear as co-authors on the cover of the corresponding volumes. As for the other active participants of the *Seminar*, whose role (as much from the point of view of writing up as from that of the work of mathematical refinement) has been growing from year to year, the name of each participant appears at the head of the exposés for which he is responsible as lecturer or as writer, and the list of those who appear in a given volume is indicated on the flyleaf of the said volume.

It is appropriate to give some details on the relations between the *Seminar* and the *Éléments*. The latter were intended in principle to give a comprehensive exposition of the notions and techniques judged most fundamental in Algebraic Geometry, as these notions and techniques themselves emerge, through the natural play of requirements of logical and aesthetic coherence. From this perspective, it was natural to regard the *Seminar* as a preliminary version of the *Éléments*, destined to be absorbed almost entirely, sooner or later, into the latter. This process had already begun to a certain extent some years ago, since Exposés I to IV of the present volume SGA 1 are entirely absorbed by EGA IV, and Exposés VI to VIII were to be absorbed within a few years in EGA VI. However, as the work of construction undertaken in the *Éléments* and in the *Seminar* develops, and as its overall proportions become clearer, the initial principle (according to which the *Seminar* would constitute only a preliminary and provisional version) appears less and less realistic because of (among other things) the limits imposed by provident nature on the length of human life. Taking into account the care generally taken in writing up the various parts of the *Seminar*, there will doubtless be reason to take up such a part again in the *Éléments* (or in treatises that might take over from them) only when progress subsequent to its writing makes it possible to bring very substantial improvements to it, at the price of fairly deep modifications. This is already the case for the present seminar SGA 1, as was said above, and also for SGA 2 (thanks to the recent results of Mme RAYNAUD). On the other hand, nothing indicates at present that this will be so in the near future for any of the parts SGA 3 to SGA 7 cited above.

References within the “Séminaire de Géométrie Algébrique du Bois Marie” are given as follows. An *internal reference* to one of the parts SGA 1 to SGA 7 of the Seminar is given in the style III 9.7, where the numeral III denotes the number of the exposé, which appears at the top of each page of the exposé in question, and 9.7 the number of the statement (or of the definition, remark, etc.) within the exposé. Where appropriate, longer decimal numbers may be used, for example 9.7.1, 9.7.2 to denote, for example, the various steps in the proof of a proposition 9.7. The reference III 9 denotes paragraph 9 of Exposé III. The number of the exposé is omitted for references internal to an exposé. For a *reference to another* of the *parts* of the *Seminar*, the same abbreviations are used, but preceded by mention of the part of the SGA in question, SGA 1 III 9.7. Likewise, the reference EGA IV 11.5.7 means: *Éléments de*

Géométrie Algébrique, Chap. IV, statement (or definition etc.) 11.5.7; here the first arabic numeral again denotes the number of the paragraph. Apart from these conventions, in force throughout the SGA, the bibliography relating to an exposé will generally be gathered at the end of it, and will be referred to within the exposé by numbers in square brackets such as [3], following the usual practice.

Finally, for the convenience of the reader, whenever it seems necessary, we shall append at the end of the volumes of the SGA an index of notation and a terminological index containing, where appropriate, an English translation of the French terms used.

I wish to add to this introduction an extra-mathematical comment. In the month of November 1969 I learned that the Institut des Hautes Études Scientifiques, of which I have been a professor essentially since its foundation, had for three years been receiving subsidies from the Ministry of the Armed Forces. Already as a beginning researcher I found extremely regrettable the lack of scruples that most scientists show in agreeing to collaborate in one form or another with the military apparatuses. My motivations at that time were essentially of a moral nature, and therefore hardly likely to be taken seriously. Today they acquire a new force and a new dimension, in view of the danger of destruction of the human species with which we are threatened by the proliferation of the military apparatuses and of the means of mass destruction that these apparatuses have at their disposal. I have explained myself elsewhere in more detail on these questions, which are much more important than the advancement of any science whatsoever (mathematics included); one may for example consult on this subject the article by G. Edwards in no. 1 of the journal *Survivre* (August 1970), summarizing a more detailed exposition of these questions that I had given elsewhere. Thus I found myself working for three years in an institution while it was taking part, without my knowledge, in a mode of financing that I consider immoral and dangerous⁵. Being at present alone in holding this opinion among my colleagues at the IHES, which has doomed to failure my efforts to obtain the removal of the military subsidies from the budget of the IHES, I have taken the decision that was called for, and am leaving the IHES on September 30, 1970, and am likewise suspending all scientific collaboration with this institution for as long as it continues to accept such subsidies. I have asked Mr. Motchane, director of the IHES, that from October 1, 1970 on the IHES refrain from distributing mathematical texts of which I am the author, or which form part of the Séminaire de Géométrie Algébrique du Bois Marie. As was said above, the distribution of this seminar will be taken over by the firm Julius Springer, in the series of the Lecture Notes. I am happy to thank here the Springer firm and Mr. K. Peters for the efficient and courteous help they have given me in making this publication possible, by taking charge in particular of the typing, for photo-offset, of the new exposés added to the old seminars, and of the missing exposés of the incomplete seminars.

I also thank Mr. J.P. Delale, who took on the thankless task of compiling the index of notation and the terminological index.

Massy, August 1970.

⁵It goes without saying that the opinion I have just expressed is my own responsibility alone, and not that of the publishing house Springer, which publishes the present volume.

Foreword

Each of the written-up exposés gives the substance of several consecutive oral exposés. It did not seem useful to specify their dates.

Exposé VII, to which reference is made at various points in Exposé VIII, has not been written up by the lecturer, who in the oral lectures had confined himself to sketching the language of descent in general categories, taking a strictly utilitarian point of view and without going into the logical difficulties raised by this language. It became apparent that a correct exposition of this language would go beyond the limits of the present notes, if only by its length. For a formal exposition of the theory of descent, I refer to an article in preparation by Jean GIRAUD. Until it appears⁶, I think that an attentive reader will have no trouble in supplying by his own means the phantom references of Exposé VIII.

Other oral exposés, coming after Exposé XI, and to which allusion is made at certain places in the text, have not been written up either, and were intended to form the substance of an Exposé XII and of an Exposé XIII. The first ones among these oral exposés took up again, in the framework of schemes and of analytic spaces with nilpotent elements (as they are introduced in the Séminaire Cartan 1960/61), the construction of the analytic space associated with a prescheme locally of finite type over a complete valued field k , the theorems of GAGA type in the case where k is the field of complex numbers, and the application to the comparison of the fundamental group defined by transcendental means and the fundamental group studied in these notes (compare A. Grothendieck, *Fondements de la Géométrie Algébrique*, Séminaire Bourbaki no. 190, page 10, December 1959). The last oral exposés sketched the generalization of the methods developed in the text to the study of coverings admitting tame ramification, and of the structure of the fundamental group of a complete curve with a finite number of points removed (compare loc. cit., no. 182, page 27, theorem 14). These exposés introduce no essentially new idea, which is why it did not seem indispensable to give a formal write-up of them before the appearance of the corresponding chapters of the *Éléments de Géométrie Algébrique*⁷.

On the other hand, the theorems of Lefschetz type for the fundamental group and the Picard group, from the local as well as from the global point of view, have been the subject of a separate Seminar in 1962, which has been completely written up and is available to users⁸. Let us point out that the results developed both in the present Seminar and in that of 1962 will be used in an essential way in the appearance of several key results in the étale cohomology of preschemes, which will be the subject of a Seminar (conducted by M. Artin and myself) in 1963/64, at present in preparation⁹.

Exposés I to IV, of an essentially local and very elementary nature, will be entirely absorbed by Chapter IV of the *Éléments de Géométrie Algébrique*, the first part of which is in press and which will doubtless be published around the end of '64. They may nevertheless be useful to a reader who would like to become acquainted with the essential properties of

⁶Now published: J. GIRAUD, *Méthodes de la descente*, Mémoire no. 2 of the Société mathématiques de France, 1964.

⁷They are included in the present volume in Exp. XII by Mme Raynaud, with a proof different from the original proof expounded in the oral Seminar (cf. introduction).

⁸*Étale cohomology of coherent sheaves and local and global Lefschetz theorems* (SGA 2), published by North Holland Pub. Co.

⁹*Étale cohomology of schemes* (cited SGA 4), to appear in this same series.

smooth, étale or flat morphisms before entering into the arcana of a systematic treatise. As for the other exposés, they will be absorbed into Chapter VIII¹⁰ of the “*Éléments*”, whose publication can hardly be envisaged for several years.

Bures, June 1963.

¹⁰In fact, as a result of a modification of the plan initially envisaged for the *Éléments*, the study of the fundamental group is postponed there to a chapter later than the one just indicated. Compare the introduction that precedes the present “Foreword”.